Clinic Appointments Analysis

Healthcare Business Intelligence & Operational Analytics Framework

Author: Ahmed Mohamed Kamel Ibrahim Abdel-Aal

Tools: Python, SQL, Power BI, Tableau, DAX, Excel

Executive Summary

This project analyzes 3,218 appointments across 1,264 patients to evaluate operational performance, patient attendance, departmental workloads, and revenue collection. Interactive dashboards were developed using Power BI and Tableau to support data-driven healthcare decision-making.

Key KPIs

- Total Appointments: 3,218
- Total Patients: 1,264
- Total Billed Amount: $10.76M
- Total Paid Amount: $5.87M
- Outstanding Balance: $4.88M
- Collection Rate: 54.61%
- Completion Rate: 49.53%
- No-Show Rate: 15.85%

Key Insights

Dermatology recorded the highest visit volume, followed by ENT and Internal Medicine. AXA and GlobeMed generated the highest appointment volumes among insurance providers. Revenue remained relatively stable throughout the year, while collection performance requires improvement.

Strategic Recommendations

- Reduce no-show rates through SMS and email reminders.
- Improve collection processes for outstanding balances.
- Increase capacity in high-demand departments.
- Enhance payer collaboration with AXA and GlobeMed.
- Develop predictive analytics for appointment forecasting.

Project Phases

Phase 1: Data Engineering using Python, SQL, and Power Query.
Phase 2: Data Modeling using Power BI and DAX.
Phase 3: Visualization using Power BI and Tableau.
Phase 4: Deployment and reporting using Excel and web technologies.

Conclusion

The Clinic Appointments Analysis project demonstrates how business intelligence tools can improve healthcare operations through better scheduling, financial monitoring, and patient management. The generated insights help increase efficiency, improve revenue collection, and enhance patient experience.